

Wave 4 series Dispersion-Aware Modeling Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the approved wave 4.1 robust-loss dispersion-aware campaign. The campaign tested whether less outlier-sensitive pointwise losses help the curve-aware harmonic residual-offset probe handle locally dispersed TE data before moving to quantile, probabilistic, mixture, latent-state, or multi-head architectures.

The campaign completed all planned entries

- 9 completed runs;
- 0 failed runs;
- 3 robust loss profiles;
- 3 required direction surfaces: global , Fw , and Bw .

The runner-level scalar first entry is te_track2h_smooth_11_robust_bw . This is not a deployment winner by itself. Wave 4 series must still pass the official curve-first TE curve verification refresh before promotion decisions.

Campaign Artifacts

Artifact	Path
Campaign output	output/training_campaigns/2026-06-11-11-51-10_track2h_dispersion_aware_modeling_campaign_2026_06_10
Leaderboard	output/training_campaigns/2026-06-11-11-51-10_track2h_dispersion_aware_modeling_campaign_2026_06_10/campaign_leaderboard.yaml
Best-run pointer	output/training_campaigns/2026-06-11-11-51-10_track2h_dispersion_aware_modeling_campaign_2026_06_10/campaign_best_run.yaml
Execution report	output/training_campaigns/2026-06-11-11-51-10_track2h_dispersion_aware_modeling_campaign_2026_06_10/campaign_execution_report.md
Planning report	doc/reports/campaign_plans/track_2/2026-06-10-16-56-36_track2h_dispersion_aware_modeling_probe_campaign_plan_report.md
Campaign package	config/training/track2h_dispersion_aware_modeling/campaigns/2026-06-10_track2h_dispersion_aware_modeling_campaign

Execution Summary

Field	Value
Campaign name	track2h_dispersion_aware_modeling_campaign_2026_06_10
Completed runs	9
Failed runs	0
Model type	curve_aware_harmonic_residual_offset_probe

Field	Value
Loss profiles	mae , smooth_l1 , log_cosh
Tested surfaces	global , Fw , Bw
Runner-level scalar first entry	te_track2h_smooth_l1_robust_bw
Program-level scalar winner changed	no

Directional Branch Results

Surface	Candidate	Loss Profile	Test MAE	Test RMSE	Val MAE
global	mae_robust_global	mae	0.003406	0.003807	0.003645
Fw	mae_robust_fw	mae	0.003146	0.003527	0.003258
Bw	smooth_l1_robust_bw	smooth_l1	0.003074	0.003662	0.003372

These rows are the branch-level closeout result. The Bw row is also the campaign scalar leader, but it does not replace the global or Fw branch.

Robust-Loss Leaderboard

Rank	Surface	Candidate	Loss Profile	Test MAE	Test RMSE	Val MAE
1	Bw	smooth_l1_robust_bw	smooth_l1	0.003074	0.003662	0.003372
2	Fw	mae_robust_fw	mae	0.003146	0.003527	0.003258
3	Fw	smooth_l1_robust_fw	smooth_l1	0.003314	0.003679	0.003235
4	Fw	log_cosh_robust_fw	log_cosh	0.003355	0.003708	0.003280
5	global	mae_robust_global	mae	0.003406	0.003807	0.003645
6	global	smooth_l1_robust_global	smooth_l1	0.003422	0.003810	0.003641
7	Bw	mae_robust_bw	mae	0.003430	0.004029	0.003579
8	Bw	log_cosh_robust_bw	log_cosh	0.003481	0.004029	0.003774
9	global	log_cosh_robust_global	log_cosh	0.003505	0.003935	0.003645

Baseline Comparison

Surface	T2H Best	T2H MAE	T2G Ref.	Ref. MAE	Delta
global	mae_robust_global	0.003406	full_curve_composite_global	0.003345	-1.82%
Fw	mae_robust_fw	0.003146	raw_centered_shape_fw	0.003181	+1.10%
Bw	smooth_l1_robust_bw	0.003074	pointwise_control_bw	0.003430	+10.38%

Scalar training metrics show a useful but direction-dependent result. Robust losses clearly improve the backward scalar branch relative to the Wave 3.3 backward reference and slightly improve the forward scalar branch. The global branch does not improve over the Wave 3.3 full-curve-composite reference.

Loss Interpretation

Loss Profile	Observed Signal	Interpretation
mae	Best global and best Fw Wave 4 series scalar result.	Absolute-error pressure is useful where local dispersion likely distorts MSE-style training.
smooth_l1	Best Bw and campaign scalar leader.	Huber-like behavior is the strongest first signal for the backward dispersion/offset problem.
log_cosh	No branch winner and weakest global result.	It is stable but not competitive in this first robust-loss campaign.

The main modeling result is not that robust losses solve TE Curve Verification Pipeline. The useful result is that robust loss selection matters, especially on the backward surface. This supports continuing the dispersion-aware plan before freezing a multi-head architecture.

Registry Effects

The campaign runner refreshed family registries for all nine Wave 4 series families and updated the program registry. The program-level scalar best did not move: te_periodic_gru_sequence_remote_Bw remains the current scalar program winner with test MAE = 0.002344 .

Registry Scope	BestRun	Test MAE
Wave 4.1 global robust	te_track2h_mae_robust_global	0.003406
Wave 4.1 Fw robust	te_track2h_mae_robust_fw	0.003146
Wave 4.1 Bw robust	te_track2h_smooth_l1_robust_bw	0.003074

TE Curve Verification Pipeline Boundary

Official TE curve-first verification was not run as part of this normal campaign closeout. Under campaign governance, that remains a separate operator-approved workflow after this campaign-results report and PDF are complete.

The next TE Curve Verification Pipeline package should add all nine Wave 4 series candidates and report accepted results separately for:

- global ;
- Fw ;
- Bw .

The verification must compare raw curve error, centered-shape error, offset, amplitude, harmonic behavior, collage plots, and overlay plots against Wave 3.3, Wave 3.2, Wave 2.2, and the current accepted TE Curve Verification Pipeline baselines.

Closeout Decision

Wave 4.1 robust-loss execution is complete: all planned candidates have successful training artifacts, no failed run remains, registries were refreshed by the runner, and the active campaign state can be cleared.

From a modeling standpoint

- carry `mae_robust_global` forward as the strongest Wave 4 series global scalar candidate;
- carry `mae_robust_fw` forward as the strongest Wave 4 series forward scalar candidate;
- carry `smooth_l1_robust_bw` forward as the strongest Wave 4 series backward scalar candidate;
- keep `log_cosh` as a completed negative/weak baseline, not the next default loss profile;
- do not promote Wave 4 series before official TE curve verification.

Recommended Follow-Up

1. Accept this closeout and clear the active campaign state.
2. Prepare a separate operator-launched TE curve verification refresh for all nine Wave 4 series candidates.
3. If the official TE Curve Verification refresh confirms the scalar backward improvement, make robust losses part of the candidate set for later multi-task / multi-head models.
4. Prepare the next Wave 4 series package in staged order: quantile/probabilistic regression, mixture-density heads, then latent-state or hysteresis-aware models.
5. Keep `Fw`, `Bw`, and `global` as parallel branch decisions.